

TAPI RIVER REJUVENATION:

RESEARCH QUESTION:

What is the need to carry out the Tapi River rejuvenation project in Surat, Gujarat?

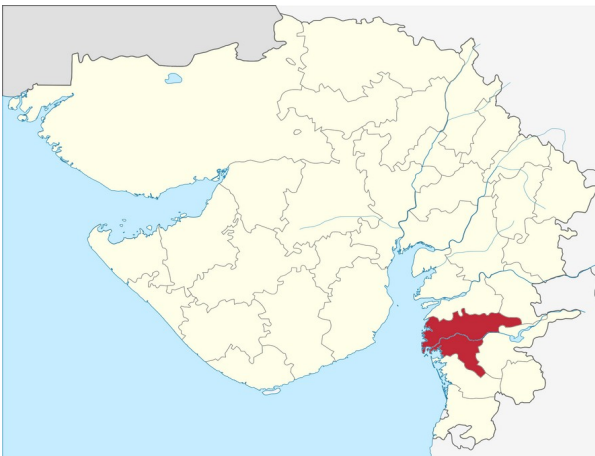
ABSTRACT:

With the rapid urban development, there are new challenges to the trends in urbanization. The current trends in the urban development have given a rise to make a daily contact with natural habitats. So, it is undoubtedly essential to consider the natural system while designing a new urban environment. This research is based on the whole idea of the Tapi River rejuvenation project and why it is so essential to conduct and work upon this project for improving the environmental quality and the entire aquatic ecosystem. In addition to it, this research will throw some light on approaching a riverfront development project and developing strategies and ideologies for natural and ecological restoration of River Tapi that is environmentally free from hazards which will be caused by the urbanization near the river front.

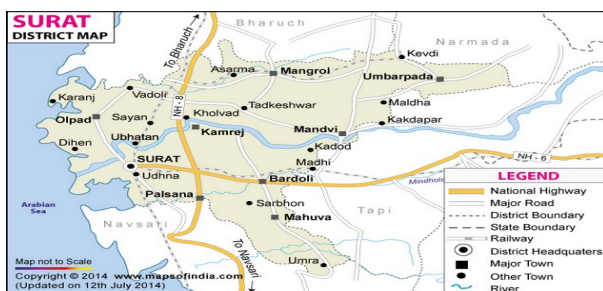
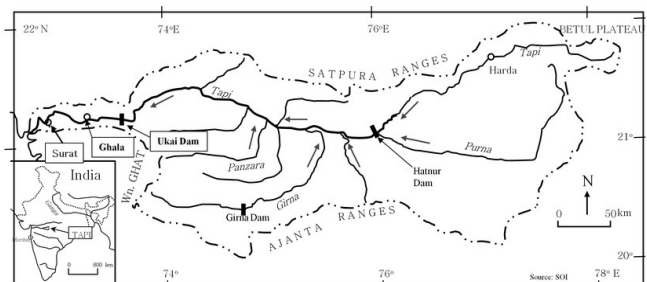
INTRODUCTION:

Surat is technologically advancing and making its size bigger day-by-day. It is upgrading itself to the newer technologies that are coming today. The research undertakes an in-depth study through the background information on the Tapi River rejuvenation project in Surat. The research will also include potential data sources as well as information and methodology for the research carried out with the Tapi River rejuvenation in Surat city.

Surat city is located in a state called Gujarat, India. It has an area of 462.15 km². This is the current area of this city, according to the recent demographics uploaded by the Municipal Corporation of Surat, Gujarat. It has a population of around 77,84,000 and with an overall growth rate of 3.93% than the previous year. Surat is the 8th largest city in India in terms of population and largest in non-capital cities. As per city mayor's foundation report, Surat is the 4th fastest growing city globally. It has a vibrant economy. It is the biggest diamond and textile hub of India. According to an estimate, Surat accounts for 90% of the world's total rough diamond cutting and polishing. In textiles, Surat contributes 40% of the nation's total manmade fabric production, 28% of the nation's total manmade fibre production. Surat is one of the cleanest city in India. It is known as the economic capital of India. It is ranked 4th under the Smart City Mission of India. In the past, this city was a glorious port with ships of more than 84 countries anchored in its harbour at any time.



- 8th largest city of India.
- 2nd largest city of Gujarat.
- Current estimate city population (2023): 61,61,000.
- Current estimate metropolitan population (2023): 63,31,000



The Tapi river basin encompasses an area of 65,145 km², which is nearly 2% of the total area of India. The basin lies in the states of Maharashtra, Madhya Pradesh and Gujarat. The river passes through rich and highly cultivated plains that form the central part of the Surat district. Occurrences of floods every monsoon form a significant feature of the Tapi estuary region. The flow in the river is regulated by Ukai multipurpose dam, which was commissioned in 1972.



The most ancient source of water, even before the human race came into being, was the rivers. The river systems have sustained civilizations since the very beginning. All the rivers and its tributaries are conventionally and dynamically the source of livelihood for the people involved in the following drinking and domestic uses, occupation/trade/tradition: agriculture, power generation, fisheries, navigation, transport and tourism, poultry and livestock farming, pottery, sand-mining, art and craft, several small and medium scale industries, such as apiculture, sericulture and jute, etc, also large scale of industries, such as leather industries, pharmaceuticals, electronics, textile, paper, tanneries, fertilizer and oil refineries.

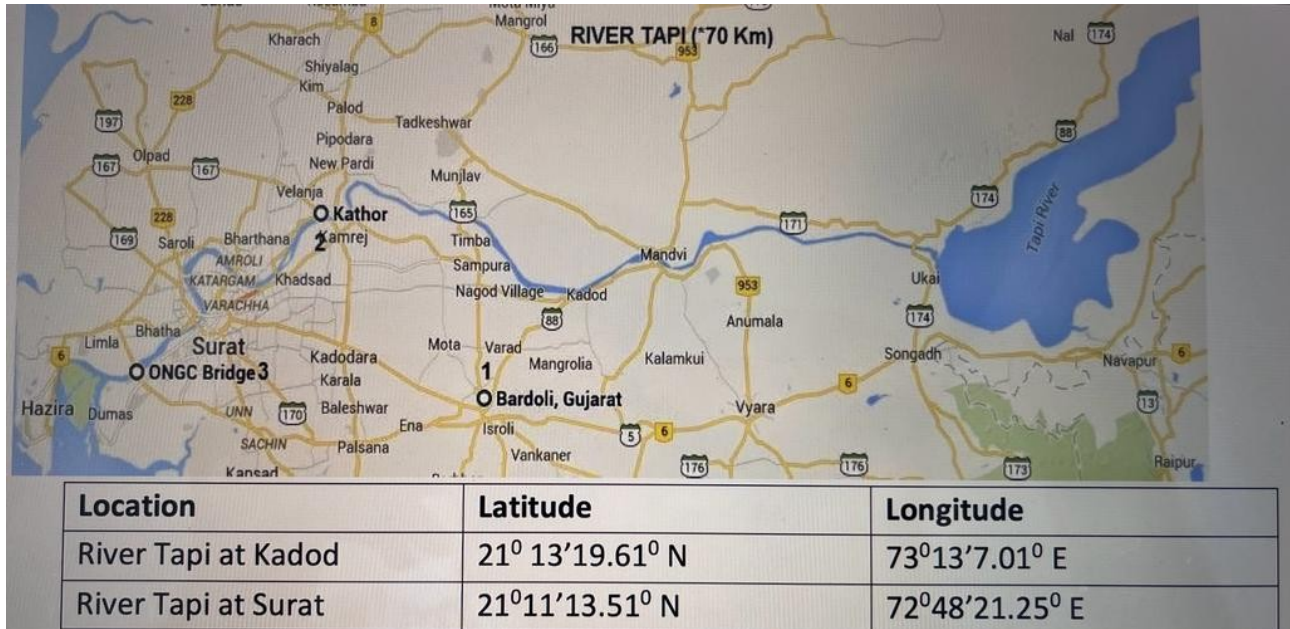
Status of Indian Rivers:

Apart from being the major of human needs and agriculture, river water is an important aspect with respect to transport, energy, habitat, civilization and industry. Discharge of partially treated and untreated sewage and discharge of industrial wastewater are major reasons of organic pollutants in rivers. A significant difference exists between the volume of municipal wastewater generation of 61,948 million litres per day against the installed sewage treatment capacity of 23,277 million litres per day.

HOW DID THE PROBLEM GET SOLVED USING GIS:

The prime objective of the work was to improve the water quality of river Tapi. The Government of India took some aerial images from the satellite to analyze on how can we answer the question of rejuvenating the Tapi River in Surat.

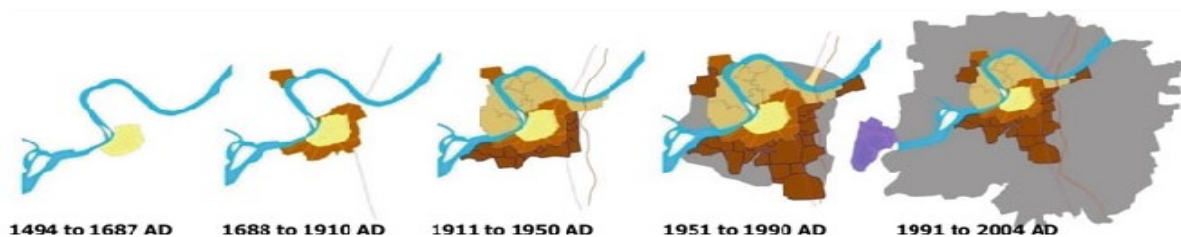
They made an action plan and found out the coordinates of the Tapi river at Kadod and Surat.



By using GIS software and technology, the government was able to locate the coordinates and was successfully able to analyze and start the redevelopment of the river Tapi for identifying key elements for cleaning of the river and making it attractive for tourists. The riverfront development and rejuvenation project was given to Surat so that the valuation of the assets in Surat become more valuable.

According to the Project masterplan, the edges on both the sides of the river, which is of 33 kms, will be constructed strong.

The images beneath were taken from the GIS Software, indicating the growth of the river from 1494 AD till 2004 AD.



The above image is taken recently from Surat Municipal Corporation demonstrating the new project River Tapi Rejuvenation in Surat.



The main rejuvenation plan of the river Tapi by the Surat Municipal Corporation looks something like the images above.

EXECUTIVE SUMMARY:

Surat city is going to be the next Chicago in terms of development. It is being technologically advancing itself gradually, looking at the developed metropolitan cities like Mumbai and Delhi. It is trying to improve its infrastructure and lifestyle of people living in the city through various advancements. The contaminated smell and gloomy landscapes have already become a reason for the public, not to engage with the current and existing riverfront walk. The Tapi rejuvenation project is conducted by the government of India, to initialise cleanliness and better standards of living for the people living in Surat. In addition, the project design considers the

This project carefully examines all the issues pertaining to river and suggests all-inclusive, immediate and long-term solutions. The Tapi Rejuvenation Project and river front development authority considers the existing situation of river in context of floods and already constructed flood embankment schemes and purposes to resolve the present issues and unlock potential opportunities. This project also intends to integrate other parallel relevant projects like the Heritage precinct redevelopment, development of riverfront parks and walkway.

Additionally, the project aims to clean the river water through aeration systems, fulfil the water requirement for which people would otherwise depend on groundwater, provide universally open spaces and strengthen the road network.

The river and its surroundings are vulnerable to floods, which also leads to discharge of contaminated sewage from storm water outfalls. Apart from the issues of water management after the construction of Singanpore weir, the river bank has also witnessed rise in squatter settlements and slums. While there have been isolated attempts to curb these issues such as providing recreational facilities on the banks of river, these efforts of development needs to be integrated in a comprehensive manner to see a truly rejuvenated Tapi River.

The concept masterplan is an attempt to formulate a blueprint of protected, feasible, legible and resilient river edge enriched with spaces designed and designated for festivals and commercial activities. The river front is proposed to facilitate public interaction with the river making it clean and accessible and to transform the existing neglected riverfront into a centre for social, cultural and recreational activities in the city.

The Tapi River rejuvenation project emphasizes can directly impact some of the key pillars of “The Surat Resilience Strategy”, for example, water availability and quality. Dominant sectors of employment and economic development, ecosystem and environmental regulation, social cohesion, upscaling of health, affordable housing and connectivity and mobility services with regulatory measures. This initiative is taken into account to develop the Tapi River (33 Km lateral range from Kathore Bridge to the proposed Rundh-Bhatha Barrage) in a holistic manner and increase its asset value for Surat.

The ballpark cost of the project is Rs. 3,904 crores and the project will be implemented in 2 phases:

- **Phase 1:**

The first phase comprises of the priority components to be executed under the head of World Bank Funds such as embankment construction, promenade finishing, toilet and lighting for entire 33 kms. This cost up to approximately Rs. 2,982 crores.

- **Phase 2:**

The other components such as Eco Park, recreational park, informal market, pedestrian bridges and other public utilities shall be executed at a later stage. The proposed Rundh Bhatha Barrage and Interceptor lines proposed under the Tapi Shudhhikaran Yojna will be integrated along with the Riverfront Development works.

